## Testosterone

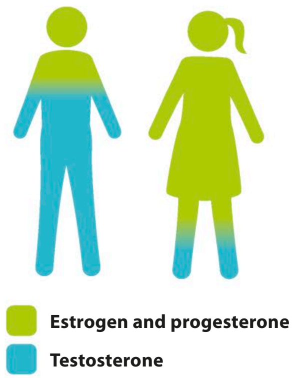
Testosterone is the game changer during puberty. It is converted in the testicles to dihydrotestosterone (DHT)—the active form of the hormone—and is regulated by low levels of progesterone.

It is DHT that causes the changes to a male's body during puberty. DHT works on target organs to help the genitals to grow, as well as to develop pubic, body, and facial hair. It also helps the prostate gland to grow, helping semen production. The amount of testosterone in the body controls the levels of DHT.

△ **Getting bigger**
Different levels of testosterone cause males to mature at different rates.

## Shared hormones

Estrogen, progesterone, and testosterone are present in both females and males, but in vastly different amounts. Throughout their lives, males produce some estrogen and progesterone and females continue to produce low levels of testosterone. In males, estrogen controls body fat and contributes to the sex drive, while progesterone monitors testosterone production. In females, testosterone is linked to maintaining bone and muscle mass, and contributes to the sex drive. Hormone levels differ between people and change over a lifetime.




△ **Hormone levels**
Males produce ten times more testosterone than females, but only half the amount of estrogen and progesterone.

## Other hormones

Hormones don't just prompt the start of puberty and the development of sexual characteristics. There are lots of different types at work in everybody, regardless of sex, that control and coordinate many bodily functions to keep the body healthy.

### Maintaining a healthy body

- Antidiuretic hormone (ADH) keeps the body's water levels balanced.
- Melatonin allows the body to sleep at night and stay awake during the day.
- Thyroxine determines how quickly or slowly the body metabolizes food.

### Managing food processes

- Leptin regulates the appetite by making the body feel full after eating.
- Gastrin triggers gastric acid in the stomach, which breaks down food.
- Insulin and glucagon control how much sugar is released into the blood after eating.

### Coping with stress

- Adrenaline raises the heart rate and produces energy when a person is under stress.
- Cortisol manages the brain's use of sugars, providing more energy.
- Oxytocin enables bonds with other people by reducing fear and creating feelings of trust.